

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

Limited access to high-quality annotations remains a broad obstacle for training robust vision models in specialized visual domains, with medical imaging serving as a particularly demanding case. Self-supervised masked image modeling (MIM) provides an annotation-efficient pretraining route, but many widely used MIM methods still depend on random, high-ratio masking that can be inefficient and weakly aligned with task semantics. Existing region-aware alternatives often rely on reconstruction heuristics or supervised localization signals, which can restrict their adaptability across datasets, modalities, and downstream objectives. We introduce Mask What Matters, a controllable text-guided masking framework for self-supervised medical image analysis. The method uses vision-language models to convert open-vocabulary prompts into region-localization cues, then applies differentiated masking that emphasizes diagnostically relevant regions while reducing redundant masking in background areas. This prompt-conditioned design supports semantic alignment during pretraining, improves representation learning, and is intended to transfer across the evaluated tasks and medical imaging modalities without changing the underlying MIM pipeline. Across modality-diverse evaluations on brain MRI, chest CT, and lung X-ray data, Mask What Matters consistently improves over existing MIM methods (e.g., SparK), with gains of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection. These improvements are obtained under the same datasets and evaluation conditions while using substantially lower overall masking ratios (e.g., 40% vs. 70%). The results indicate that controllable, text-driven masking is a broadly applicable mechanism for semantically aligned self-supervised learning in the studied medical imaging settings, offering a practical path toward more robust vision models when annotations are limited.

1 INTRODUCTION

Image classification, segmentation, and object detection form a common foundation for visual recognition systems across many application domains. Deep learning has become the dominant technical driver for these tasks, but its strongest performance usually depends on large annotated datasets. In specialized settings such as medical image analysis, annotations are often costly, scarce, and dependent on expert knowledge, creating a recurring barrier to deploying high-capacity vision models beyond well-labeled benchmark environments.

Self-supervised learning (SSL) addresses this limitation by using unlabeled data for pretraining, allowing models to acquire transferable representations before task-specific fine-tuning (Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)). Within SSL, MIM has become a broadly used approach: parts of an image are masked, and the model learns to reconstruct the missing content from visible context. This objective encourages the capture of structural and semantic regularities, has performed well across downstream vision tasks, and has been adapted to medical image analysis (He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)).

However, common MIM methods, including masked autoencoders (MAE) [He et al., 2022] and SparK [Tian et al., 2023], generally use random masking strategies that are not explicitly aligned with regions that matter for downstream decisions. This task-agnostic design creates two related issues: (1) learned representations can become weakly aligned with downstream semantics, reducing transfer performance when the relevant visual evidence is spatially concentrated; and (2) to increase the likelihood that informative areas are masked, existing image-based methods often use very high masking ratios (e.g., 75% in MAE), which raises reconstruction difficulty and can require substantial data and computation. By contrast, natural language has compact structure and high semantic density, supporting effective masked learning even at low masking ratios [Devlin et al., 2019]. The need for high masking ratios in images is therefore partly a consequence of visual redundancy and the absence of semantic-aware masking, rather than an inherent requirement for all MIM settings.

These challenges are especially visible in medical imaging, but they also reflect a broader class of specialized visual tasks in which useful evidence may occupy only a small part of the image. In the medical setting studied here, diagnostically relevant information is often localized to lesions, organs, or anatomical subregions, while large areas may contribute little to a given objective. Medical imaging also spans modalities such as MRI, CT, and X-ray, where the size, shape, appearance, and spatial distribution of task-relevant regions vary substantially across datasets and tasks. These properties make efficiency, controllability, and cross-setting adaptability important requirements for masking strategies.

To address this need, we propose a controllable, text-guided masking framework, **Mask-What-Matters (MWM)**. Given a user-specified open-vocabulary phrase or description, a pretrained vision-language model provides zero-shot localization cues that are converted into robust regions of interest (ROIs). MWM then applies prompt-conditioned, region-specific masking, using higher masking ratios for semantically important areas such as lesions or organs and lower ratios for background. In this way, downstream-relevant semantics are introduced into the pretraining signal without requiring per-image reports or labels.

To summarize, our main contributions are:

- A controllable text-guided masking framework for medical imaging. MWM integrates vision-language models (BiomedCLIP) with segmentation refinement (SAM) to localize task-relevant regions from open-vocabulary prompts and apply differentiated ROI vs. background masking, addressing the semantic misalignment of random masking.
- An annotation-free and backbone-agnostic design. MWM requires no per-image reports or labels and can be integrated into both ViT- and ConvNet-based MIM pipelines under a unified protocol.
- Consistent gains across the evaluated modalities and tasks. On brain MRI, chest CT, and X-ray datasets, MWM surpasses state-of-the-art MIM baselines (e.g., SparK), improving classification, detection, and segmentation performance and demonstrating the promise of text-driven generative pretraining for medical image analysis.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling was first popularized in natural language processing through bidirectional encoder representations from transformers (BERT) [Devlin et al., 2019], which learns contextual representations by masking tokens and predicting them during pretraining. This principle was later transferred to computer vision by [He et al., 2022], where image regions are hidden and the model reconstructs the missing content. The resulting MIM paradigm has shown strong transfer behavior in both natural-image and medical-image settings [Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023]. Representative methods include MAE [He et al., 2022], MedMAE [Gupta et al., 2025; Xiao et al., 2023], and SparK [Tian et al., 2023]. MAE uses high-ratio random masking to encourage ViTs to learn high-level semantic representations; MedMAE adapts this idea to medical images; and SparK extends MIM to convolutional backbones by using their multi-scale feature structure for efficient pretraining.

Despite these advances, most conventional MIM approaches use random or structure-agnostic masking, which does not distinguish between regions that are central or peripheral to downstream ob-

jectives. Such non-selective masking can encourage the model to spend capacity on irrelevant or redundant image content, limiting transfer in settings where semantic evidence is localized. Recent region-aware masking methods attempt to improve this alignment, and two main directions have emerged:

One direction uses loss-driven adaptive masking. Methods such as hard patches mining (HPM) [Wang et al., 2023a] and AnatoMask [Li et al., 2024] estimate reconstruction difficulty across regions and assign higher masking probabilities to areas with larger losses, with the goal of emphasizing visually informative structures. However, reconstruction difficulty is not always equivalent to downstream relevance. In brain magnetic resonance imaging (MRI), for example, visually complex structures such as the spinal cord or eyeballs may receive high attention even when they are not central to tasks such as tumor detection.

Another direction uses external perceptual modules to guide masking. FocusMAE [Basu et al., 2024], for instance, relies on pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. This can be effective when suitable region proposal supervision is available, but the need for supervised, domain-specific RPN training limits portability across tasks and datasets.

These limitations motivate masking mechanisms that are more flexible, semantically aligned, and transferable across specialized visual settings. Natural language offers a compact and interpretable source of task guidance, making it a promising way to steer masked modeling without requiring dense labels or per-image supervision.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Recent progress in vision-language alignment has made it increasingly feasible to identify semantically meaningful visual regions from text under zero-shot and open-vocabulary settings [Radford et al., 2021]. The Contrastive Language-Image Pretraining (CLIP) family learns a shared embedding space for images and texts through large-scale image-text pretraining, enabling broad use in zero-shot and open-set recognition. In medical imaging, models such as BiomedCLIP [Zhang et al., 2023] adapt this cross-modal alignment to biomedical data and support downstream applications across multiple medical tasks. Although vision-language pretrained models are now widely used in supervised or downstream pipelines, much of this work focuses on applications such as object detection and semantic segmentation [Zhong et al., 2022, Aleem et al., 2024, Koleilat et al., 2024]. Their open-vocabulary and zero-shot localization capabilities remain underexplored as a mechanism for shaping the masking process during MIM pretraining.

In summary, recent region-aware MIM methods based on reconstruction loss or external perception modules improve semantic focus but still face limitations: additional supervision requirements, possible mismatch between reconstruction difficulty and task semantics, and limited flexibility across settings. At the same time, vision-language models provide a general source of prompt-conditioned semantic guidance that has rarely been integrated into the MIM pretraining stage. These observations motivate using vision-language signals to make masked image modeling more efficient and more adaptable across the evaluated medical imaging scenarios.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** is organized around two reusable stages: (1) text-guided region localization and (2) region-aware masked image modeling. The localization stage uses a frozen vision-language model to identify regions associated with an open-vocabulary prompt, and the masked modeling stage applies differentiated masking ratios across these regions to guide reconstruction toward task-relevant content.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a multi-stage pipeline designed to operate from text rather than per-image annotations. The process begins with LLM-based prompt generation and uses

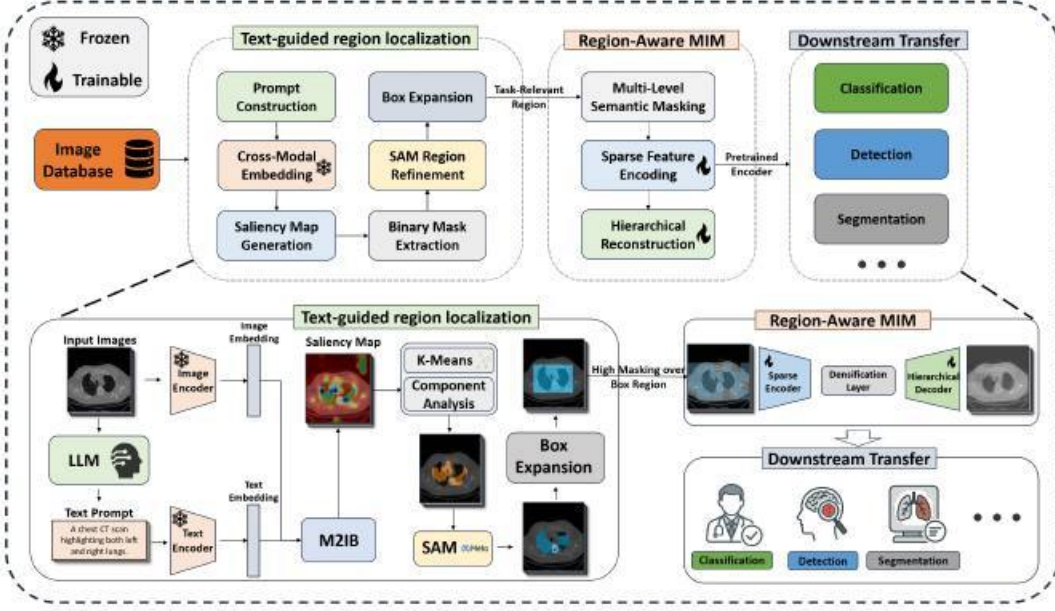


Figure 1: Overview of the MWM framework. The top panel presents the three-stage pipeline used across the study: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel illustrates how open-vocabulary prompts guide both region localization and masking.

BiomedCLIP to extract cross-modal embeddings. A saliency map is produced with the Multi-Modal Information Bottleneck (M2IB), binarized with K-Means, and refined with the Segment Anything Model (SAM). The refined mask is then converted into a bounding box with a controllable expansion margin, providing a robust spatial guide for the subsequent masked modeling stage.

Prompt Generation. Text prompts define the semantic target for localization and therefore determine how broadly the masking policy can be adapted to different categories and modalities. To provide semantic guidance, large language models (LLMs) are queried with structured requests such as “Describe the typical visual characteristics of a [category] in a [modality] image.” We also evaluate simple category phrases as alternative inputs, enabling a controlled comparison between concise and descriptive prompt styles in the experimental section.

Cross-Modal Embedding. After a prompt is specified, BiomedCLIP [Zhang et al. \(2023\)](#) encodes the medical image and the prompt into a shared semantic space. Given an image I and a prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These cross-modal embeddings provide the semantic basis for estimating prompt-relevant saliency in the following stages.

Saliency Map Generation. A cross-modal saliency estimation module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. \(2023b\)](#) is used to localize prompt-relevant visual regions by aligning image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns an importance weight to each spatial location according to its semantic relevance to the prompt. M2IB estimates this map by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are image and text embeddings encoded by BiomedCLIP respectively, and γ controls the trade-off between preserving cross-modal relevance and filtering task-irrelevant visual information.

Binary Mask Extraction. After obtaining the saliency map, we binarize it with unsupervised K-Means clustering [Lloyd \(1982\)](#) to produce a preliminary foreground mask guided by the text prompt.

The map is clustered into two pixel groups, and the group with higher saliency values is treated as foreground. To suppress false activations and improve robustness across anatomical layouts, connected component analysis is applied to the binary map, retaining the largest foreground regions and focusing the mask on core semantic structures.

Region Refinement. To improve spatial accuracy without introducing task-specific labels, SAM is used as a refinement module [Kirillov et al. (2023)]. For each selected connected region $c_i \in \mathcal{C}^*$, its minimal enclosing bounding box is computed and used as the visual prompt for SAM.

Given the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} provides a more faithful spatial structure for region-guided pretraining.

Box-Based Region Expansion. The SAM mask obtained from the previous stage can still contain boundary noise that may distract representation learning. To make the guidance more stable across datasets and modalities, the refined SAM mask is converted into a bounding box with a controllable expansion margin, which serves as the final region of interest (ROI) for downstream masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest generated by the localization module in Section 3.2, MWM performs masked image modeling through a three-stage pipeline that can be applied to different medical imaging inputs: (1) a multi-level masking strategy that applies differentiated masking ratios based on region importance, (2) a sparse encoder that processes only unmasked patches, and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM adopts a multi-level masking strategy that uses the semantic regions identified by text-driven localization. Patches inside prompt-highlighted regions, such as areas associated with tumors or organs, are masked at a high ratio so that reconstruction emphasizes task-relevant semantic features. Background regions are masked lightly or left unmasked to reduce redundant computation and preserve contextual structure. Figure 2 provides a visual comparison of the resulting masking patterns.

Sparse Feature Encoding. Visible patches are collected into a sparse image and encoded following the SparK design [Tian et al. (2023)], allowing the same masking policy to operate through a sparse reconstruction pipeline:

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are multi-scale features. Although our implementation instantiates MWM with convolutional backbones, the interface is compatible with ViTs through token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image in a top-down manner from sparse multi-scale features. At each scale l , masked locations are filled with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

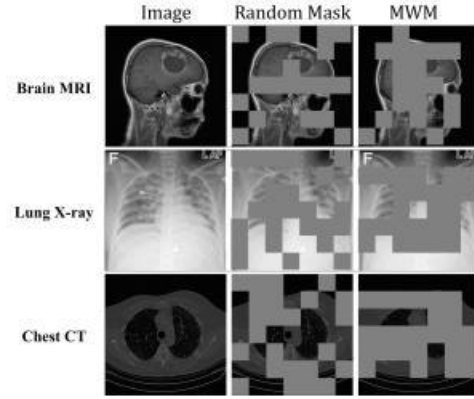


Figure 2: Comparison of masking strategies. Gray blocks indicate masked regions, illustrating how MWM concentrates masking on prompt-relevant areas while retaining more background context.

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is trained with mean-squared error on the masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i is the original patch, and $\hat{x}_i$ is its reconstruction.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. We use three self-supervised pretraining datasets that span distinct medical imaging modalities and visual structures. (1) *Brain MRI* contains approximately 17,000 images from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering multiple tumor types and anatomical structures. (2) *Lung X-ray* contains around 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021)]; Chowdhury et al. (2020), including COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* contains 12,000 slices from a lung disease dataset [Konya (2020)], with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. For downstream evaluation, we use three datasets covering classification and detection-oriented transfer. (1) *Brain Tumor MRI* [Feltrin (2023)] contains 4,479 MRI images across 44 tumor types, including astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermany et al. (2018)] contains 5,863 pediatric chest X-rays labeled as normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] contains CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

All self-supervised methods are evaluated with comparable model scales: convolution-based methods use ResNet-50 [He et al. (2016)], and transformer-based methods use ViT-S [Dosovitskiy et al. (2020)]. The input resolution is 224×224, and all models are initialized with ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} ; downstream fine-tuning keeps the same encoder and input resolution. Experiments are run on NVIDIA Tesla A100 cards with 80 GB VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, non-target variables such as prompt type or SAM usage are fixed to their empirically optimal settings so that comparisons isolate the intended experimental factor.

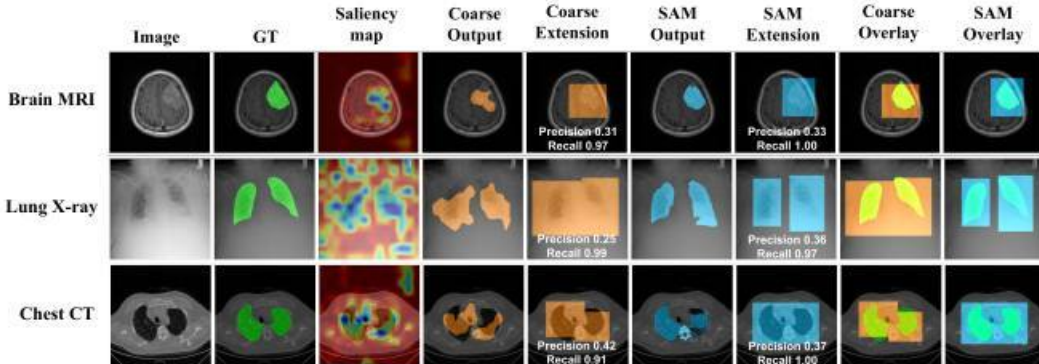


Figure 3: Representative text-guided region localization results across the evaluated medical imaging settings.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance (Occlusion Precision/Recall) across prompt types and refinement strategies in the evaluated modalities. *Coarse Ext.* denotes bounding box extension of the coarse region; + *SAM Ext.* indicates further refinement using SAM.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first evaluate whether the text-guided localization module can provide usable spatial guidance across the three studied modalities: brain MRI, lung X-ray, and chest CT. We also examine how prompt type and SAM refinement affect localization quality. Predicted bounding boxes are compared with expert-annotated segmentation masks using occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p is the predicted region and R_{gt} is the ground-truth annotation.

As shown in Table 1, text-guided localization with effective prompts achieves consistently high occlusion recall across the datasets (around 0.82) while maintaining reasonable precision. This supports the use of textual descriptions as a general spatial guide for the studied medical images, despite modality-specific anatomical variation (see Figure 3 for visualized examples).

Effect of Prompt Type on Region Localization. To assess how language granularity affects localization, we compare two prompt types. As shown in Table 1, descriptive sentence prompts perform better for Brain MRI and Chest CT, suggesting that richer semantic descriptions can help when relevant structures are more variable or localized. In lung radiographs, concise phrases achieve higher recall (0.82 vs. 0.66), likely because the lungs cover a broad anatomical area and can be captured by general category terms.

Effect of SAM Refinement on Region Localization. We further evaluate the contribution of spatial refinement by applying SAM to the coarse masks. As shown in Table 1, SAM refinement improves precision while preserving high recall. In Brain MRI, for example, precision increases from 0.22 to 0.30 while recall remains above 0.94, indicating that SAM can refine prompt-derived localization without sacrificing coverage of the target region.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next evaluate the transferability and representation quality of MWM by comparing it with representative SSL methods from both reconstruction-based and contrastive learning paradigms. All methods are pretrained and evaluated with consistent datasets and training settings, so the comparison reflects the effect of the pretraining strategy rather than differences in evaluation protocol.

Image Classification. We examine classification performance on three downstream medical imaging datasets to test whether the learned representations transfer across modality and label-space changes. Following common practice, results are reported under two settings: (1) *full fine-tuning*, where the entire model is updated; and (2) *linear probing*, where the encoder is frozen and only a linear classifier is trained on top.

- **Full fine-tuning:** Table 2 summarizes classification accuracy for MWM and several representative self-supervised methods, including MAE He et al. (2022), SparK Tian et al. (2023), AnatoMask Li et al. (2024), SimCLR Chen et al. (2020a), MoCoV2 Chen et al. (2020b), and BYOL Grill et al. (2020). MWM achieves the highest performance across the evaluated datasets, indicating that prompt-guided masking improves transfer across diverse medical imaging modalities. The largest gain appears on the Chest CT dataset, where MWM reaches 97.5% accuracy,

surpassing SparK by +3.1 and AnatoMask by +1.7. This improvement highlights the value of text-guided masking for settings where semantically important regions can be spatially sparse, such as lung lobes in CT scans, while reducing redundant masking elsewhere.

- **Linear probing:** To further evaluate representation quality independent of full encoder adaptation, we conduct linear probing experiments. For simplicity, MWM is compared with two recent state-of-the-art masked image modeling methods, SparK and AnatoMask. As shown in Table 3, MWM outperforms both across all datasets, indicating that text-driven pretraining yields features that remain useful when the encoder is frozen.

Object Detection and Instance Segmentation. To assess whether the learned representations transfer beyond classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset Hamada (2025). As shown in Table 4, MWM achieves the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), surpassing SparK by +1.3 and +1.1 points, respectively. These results show that the benefits of text-guided masking extend to object-level localization and pixel-wise segmentation under the evaluated detection setting.

Overall, incorporating text-driven masking during pretraining helps the model learn representations that transfer across the evaluated task types. The consistent gains in classification, detection, and segmentation indicate that semantic guidance can improve visual representation learning beyond a single downstream objective.

4.5 ABLATION STUDY

We conduct ablation experiments to isolate how prompt design, SAM-based region refinement, and masking ratio affect downstream classification performance in MWM. These studies examine whether the framework’s gains depend on a single configuration or persist across meaningful design choices.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As shown in Table 5, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, while the opposite pattern appears on lung X-ray. This matches the localization results in Section 4.3: prompt forms that better identify relevant regions also tend to improve downstream performance. Importantly, both prompt types, despite their different granularity, consistently outperform random masking across all datasets. This indicates that text-guided masking can align pretraining with downstream semantics across the evaluated modalities, rather than depending on a single prompt style. The SAM ablation further shows that incorporating this refinement module during pretraining consistently improves downstream classification, supporting its role in making text-guided masking spatially more reliable.

4.5.2 EFFECT OF MASKING RATIO

As shown in Figure 4, MWM achieves 96.8% classification accuracy at a masking ratio of 40%, which is substantially

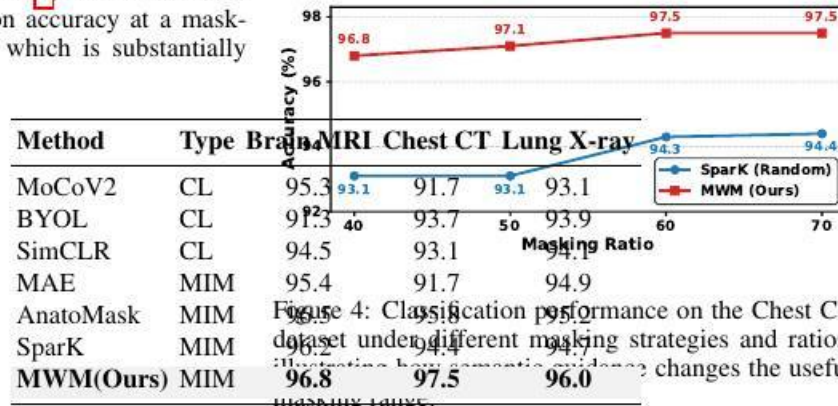


Table 2: Fine-tuning classification accuracy (%) on downstream datasets across the evaluated modalities, where MIM stands for Masked Image Modeling and CL stands for Contrastive Learning.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on downstream datasets with the encoder kept frozen.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) transfer results (%) on the BR35H dataset.

lower than the roughly 70% ratio often used in masked image modeling. Even at this lower ratio, it exceeds SparK’s best performance at 70% masking by +2.4% absolute accuracy. To ensure that SparK’s lower performance is not caused by an unfavorable masking configuration, SparK is additionally evaluated under 40%, 50%, 60%, and 70% masking ratios. The results confirm that 70% masking gives SparK its highest accuracy, supporting the robustness of the comparison. These results suggest that high masking ratios are not universally optimal for MIM. In the evaluated setting, semantically guided and task-aware masking enables stronger downstream performance at lower masking ratios by preserving relevant content and reducing redundant reconstruction burden.

5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that combines text-guided semantic localization with region-aware masking. By using natural language prompts to highlight task-relevant areas and applying differentiated masking, MWM supports semantically aligned representation learning without requiring per-image annotations.

The experiments demonstrate the effectiveness and transferability of MWM within the evaluated medical imaging scope: (1) natural language prompts provide reliable semantic region localization, supporting the feasibility of text-driven masking; (2) MWM consistently outperforms existing methods (e.g., SparK, AnatoMask) in classification across three imaging modalities and also improves detection and instance segmentation metrics; (3) MWM maintains strong performance at lower masking ratios (e.g., 40%), indicating that semantic guidance can reduce reliance on high-ratio random masking during self-supervised pretraining.

Limitations. Despite these results, several limitations remain. First, although text prompts provide flexible guidance, we have not systematically tested robustness to broader variations in prompt style, vocabulary, or noise. Second, the framework depends on externally provided textual descriptions, which may reduce reliability in application scenarios where prompts are unavailable, inconsistent, or poorly matched to the target anatomy or pathology.

Looking forward, text-driven, region-aware masking offers a general principle for making self-supervised learning more semantically grounded. Extending MWM beyond the medical imaging

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under different prompt types and region refinement strategies across the three evaluated imaging modalities. *No Prompt* refers to random masking without text-guided localization. *w/* and *w/o* denote with and without SAM, respectively.

settings evaluated here may open additional opportunities for vision–language-guided pretraining in other specialized visual domains.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

LLMs were used *only* for language editing and presentation polishing, including grammar, phrasing, and minor stylistic clarification of author-written text. They were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.